Landing Lens Based Smart Hand Gesture Recognition

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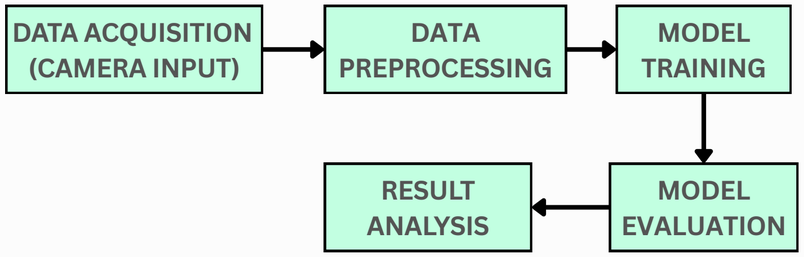
1. Abstract

This project presents a Hand Gesture Recognition system integrated with the Landing Lens platform, enabling real-time, accurate interpretation of gestures like Good Job, Dislike, Victory, Good Luck, Call Me, Okay, and Stop. A dataset of labeled images was collected using a webcam, and a model was trained to achieve reliable gesture recognition. The project aims to provide a natural, touchless interface that enhances accessibility, improves user engagement, and simplifies interaction with digital systems.

2. Introduction

Hand gesture recognition, an application of Computer Vision and Human-Computer Interaction (HCI), enables natural, touchless communication with digital systems, improving accessibility and user experience. This project integrates a robust gesture recognition module within the Landing Lens framework to deliver real-time, accurate interpretation of gestures with minimal response delay.

3. Methodology



4. Input / Output

Test Image	Actual	Predicted	Result
	Victory	Victory	✓
	Dislike	Dislike	✓
	Good Job	Dislike	✗
	Victory	Victory	✓

5. Result

The system demonstrated robust real-time performance, achieving 83% accuracy across multiple gesture classes, affirming its suitability for interactive applications requiring swift and reliable gesture interpretation.

6. Conclusion

Landing Lens hand gesture recognition enables real-time, accurate interpretation of gestures, with applications in sign language and smart home systems, and scope for expanded gestures and depth sensing.

7. Steps to use the Application

Step 1 : Take a Sample Photo of a hand gesture like the ones shown below.

Step 2: Scan the QR code.

Step 3: Upload the picture from gallery (or Take a live photo)

Step 4: Click "Use Photo" and get your result.

8. References

1. https://landing.ai/
2. https://www.google.com/
3. Bazarevsky et al. (2020) proposed a real-time hand tracking method detecting 21 3D hand landmarks from RGB images, enabling efficient gesture recognition on-device.
4. Liu et al. (2020) developed a deep learning-based hand gesture recognition system using extracted hand landmarks, achieving high accuracy and robustness.